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SCHOOL: OF SCIENCE

PROJECT TITLE

MACHO YA RAIA

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Declaration

We hereby declare that this project report, titled **macho ya raia** , is our own original work and has not been submitted, in whole or in part, for any other degree or professional qualification. All sources used or quoted have been duly acknowledged by means of complete references.

Date: March 31, 2026

DEDICATION

This project we dedicate it to each member of our development team above and the supervisor of this project who has been of great unwavering support throughout the development journey for this project .

Acknowledgement

We would like to express our deepest gratitude to our supervisors and instructors for their invaluable guidance throughout the development of this project. Their technical insights into geospatial data and software architecture were instrumental in shaping the final product.

We also wish to thank my peers and our lecturers who participated in initial surveys and feedback sessions. Your input helped ensure that **macho ya raia** was designed with the end-user in mind. Finally, we are grateful to our family and friends for their unwavering support and encouragement.

Abstract

Rapid urbanization often leads to a disconnect between citizens and local authorities regarding infrastructure maintenance and public services. **Macho ya raia** is a map-based civic engagement platform designed to bridge this gap. This project report details the development of a mobile and web-based application that allows residents to report non-emergency civic issues—such as potholes, broken streetlights, or illegal dumping—using real-time GPS tagging and photo documentation. By utilizing a full-stack architecture featuring a robust database and a responsive map interface, the system categorizes reports and visualizes them for both public awareness and administrative action. The goal of macho ya raia is to foster transparency, improve governmental accountability, and empower communities to take an active role in urban upkeep.

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<u>scope</u>

The MACHO ya raia Platform is designed to provide a real-time, map-centric interface for community reporting and civic engagement. The goal is to move beyond static reporting by creating a social, interactive "Activity Feed" synced

with geographic data, ensuring local issues and events are visible, actionable, and trackable.

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2. Functional Requirements (In-Scope)

The following features are the core pillars of the MACHO development cycle:

2.1 User Authentication & Profile Management

- **Secure Onboarding: Implementation of Google OAuth and Email/Password authentication via Supabase Auth.**
- **User Profiles: Customizable user profiles featuring activity history (reports submitted), follower/following counts, and personalized settings.**
- **Middleware Protection: Route guards to ensure sensitive pages (Dashboard, Settings, Submit) are restricted to authenticated users.**

2.2 Geospatial Reporting System

- **Coordinate Selection: A "Map Picker" tool allowing users to precisely drop a pin for a report.**
  - **Dual-Engine Map View: * Google Maps: For high-fidelity data, business searching, and detailed satellite imagery.**
    - **Leaflet: An open-source fallback for lightweight rendering and performance optimization.**
- **Report Metadata: Support for descriptions, categories (e.g., Hazard, Event, Maintenance), and timestamping.**

2.3 Community Engagement (The Social Feed)

- **Activity Feed: A real-time, scrollable feed of reports with infinite scroll capability.**

- Interactions: Users can "Like" reports and participate in threaded "Comments" for community discussion.
- Search & Filter: Capability to search for specific reports or filter by category and location.

2.4 Offline & Performance Features

- PWA Integration: Service workers to allow the app to be installed on mobile devices and desktops.
- Offline Queue: A synchronization engine (offlineQueue.ts) that caches user actions (like submitting a report) while offline and pushes them to Supabase once a connection is restored.

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3. Technical Constraints & Non-Functional Requirements

- Development Environment: The project must remain compatible with Kali Linux and the Intel i7 vPro hardware configuration.
- Type Safety: 100% TypeScript coverage to prevent runtime errors in the Ethereum-style state transitions or database calls.
- Database: Utilization of PostgreSQL via Supabase with Row-Level Security (RLS) enabled for data privacy.
- UI/UX: Responsive design using Tailwind CSS, ensuring the "BottomNav" and "Shell" layouts work seamlessly on mobile screens.

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4. Out-of-Scope (Exclusions)

To ensure the project meets its March 26, 2026, delivery goals, the following are not included in the current version:

- **Native Mobile Apps:** No iOS (Swift) or Android (Kotlin) versions; the PWA serves all mobile needs.
- **Advanced Video Processing:** While images are supported, high-definition video uploads for reports are excluded to save bandwidth.
- **Real-time Voice Chat:** Interaction is limited to text-based comments and status updates.
- **Monetization Engine:** No integrated payment gateways or subscription models are planned for this phase.

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5. Project Deliverables

1. **Source Code:** Fully documented Next.js repository.
2. **Database Schema:** Finalized SQL scripts (supabase-schema.sql).
3. **Documentation Suite:** * Setup guides for Database and Google Maps.
   - API connection documentation.
   - Final Project Status Report.
4. **Deployment-Ready Build:** A production-optimized build capable of being hosted on Vercel or a similar provider.

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6. Success Criteria

- **Latency:** Map markers must load in under 2 seconds on a standard 4G connection.
- **Sync Accuracy:** 100% of offline queued reports must successfully sync to the database upon reconnection.

- **Security: Zero unauthenticated access to the /submit or /settings routes.**

Would you like me to expand on the specific technical ar

Chapter 1: Introduction

1.1 Background Information

In the modern digital era, the concept of a "Smart City" relies heavily on the flow of information between a city's infrastructure and its inhabitants. However, many municipal feedback loops remain antiquated, relying on phone calls or physical visits to government offices. Civic technology, or "Civic Tech," seeks to leverage software to enhance the relationship between the public and the state. **Macho ya raia** emerges as a solution within this domain, focusing on crowdsourced data to identify and resolve local infrastructure failures.

1.2 Problem Statement

Currently, many citizens feel disempowered when encountering local issues like damaged public property or environmental hazards. The barriers to reporting these issues are high:

- **Lack of Centralization:** Uncertainty about which department handles specific problems.
- **Lack of Transparency:** Citizens rarely receive updates on whether their report was seen or acted upon.

- **Geospatial Inaccuracy:** Verbal descriptions of locations are often imprecise, leading to delays in maintenance. Without a streamlined, visual, and public-facing system, minor repairs escalate into costly infrastructure failures, and civic trust erodes.

1.3 Objectives

The primary objectives of macho ya raia project are:

- **To develop a user-friendly map interface** that allows citizens to drop pins on specific locations where issues occur.
- **To implement a categorization system** (e.g., Sanitation, Roads, Utilities) to ensure reports reach the correct authorities.
- **To provide a public dashboard** where users can track the status of reported issues (e.g., Pending, In Progress, Resolved).
- **To facilitate data-driven decision-making** for local governments by highlighting "hotspots" of recurring issues.

1.4 Justifications

Traditional reporting methods are inefficient and labor-intensive for both the public and the government. Macho ya raia justifies its development through:

- **Efficiency:** Automated tagging and categorization reduce administrative overhead.
- **Accountability:** Publicly visible reports encourage authorities to resolve issues timely.
- **Community Empowerment:** It transforms residents from passive observers into active stakeholders in their neighborhood’s well-being.

Chapter 2: Literature Review

2.1 Evolution of Civic Technology

Civic tech has transitioned from simple informational websites to "Web 2.0" platforms that prioritize user-generated content. Early efforts focused on transparency (open data); modern efforts focus on "actionable" data—where user input directly triggers a service response.

2.2 Review of Existing Systems

Platforms like *FixMyStreet* in the UK and *SeeClickFix* in the US have set the standard for civic reporting. Research indicates that these platforms succeed because they reduce the "friction of participation." However, many international versions lack localization for specific regional infrastructures or fail to integrate with local database systems, creating a need for a bespoke solution like macho ya raia aka citizen eyes .

2.3 Geospatial Integration in Public Service

The use of Geographic Information Systems (GIS) is no longer restricted to urban planners. By integrating Google Maps or OpenStreetMap APIs, macho ya raia allows for "Spatial Crowdsourcing." Literature suggests that visually represented data is more easily interpreted by human managers than long lists of text-based complaints.

2.4 The Role of Crowdsourcing in Governance

Crowdsourcing serves as a "distributed sensor network." Instead of a city hiring thousands of inspectors to check streetlights, every citizen with a smartphone becomes a potential inspector. This model significantly lowers the cost of urban monitoring and increases the speed of incident detection.

Additional Issues & Considerations

- **Data Privacy:** The system must ensure that while reports are public, the personal identities of the reporters are protected according to local data protection laws.
- **Verification Mechanisms:** To prevent "spam" or false reporting, the project will explore the use of community voting (upvoting an issue to confirm it exists) and AI-based image verification.
- **Scalability:** The architecture must be capable of handling increased traffic during weather-related emergencies or large-scale infrastructure failures.